THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 12	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$: 3+4+1+4=12.
2	**(c)** increases 100 times	Each unit of pH is a factor of 10.
3	**(d)** silver	Silver lies below hydrogen.
4	**(a)** 13	3n + 1 = 13.
5	**(c)** Fe_2O_3	Fe_2O_3 loses oxygen.
6	**(a)** anhydrous sodium carbonate	Loses water of crystallisation.
7	**(d)** They have high melting points	Strong forces between ions.
8	**(a)** $CH_2{=}CH_2$	Dehydration.
9	**(a)** $FeSO_4$ (pale green) is formed	Iron displaces copper.
10	**(d)** neutral	No colour change shows neutral.
11	**(c)** keeps air and water away	Barrier effect.
12	**(c)** forms ionic bonds with chlorine	Carbon forms covalent bonds.
13	**(a)** 5	$C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$.
14	**(a)** pH 1	Lower pH means more H^+.
15	**(b)** sodium	Only highly reactive metals react with cold water.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Physical change: no new substance, e.g. melting of ice (1); Chemical change: new substance formed, e.g. rusting of iron (1).	1+1
17	Strong acid ionises completely, e.g. HCl (1); Weak acid ionises partially, e.g. CH_3COOH (1).	1+1
18	Bubbles of hydrogen stick to its surface (1); They make it lighter than water (1).	1+1
19	Needs 4 electrons to complete octet (1); Loss/gain requires too much energy (1).	1+1
20	$2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Hydrogen gas (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Al on LHS 1, RHS 2: put 2 before Al (1); $2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Displacement reaction (1).	1+1+1
22	HCl: vigorous bubbling (1); CH_3COOH: slow bubbles (1); Water: no reaction; HCl is a stronger acid (1).	1+1+1
23	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (1); A thin layer of Al_2O_3 forms on the surface (1); It prevents further reaction (anodising thickens it) (1).	1+1+1
24	Formula 3n + 1 (1); Hexane: 19 (1); Decane: 31 (1).	1+1+1
	OR: Reaction is endothermic (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); Quick lime is used in cement/whitewash (1).	1+1+1
25	Electrolysis of brine (1); Anode: Cl_2; cathode: H_2 (1); NaOH remains in solution (1).	1+1+1
	OR: Crushing and grinding (1); Removal of gangue (1); Roasting converts ZnS to ZnO (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(a) Displacement (1); (b) Decomposition (1); (c) Double displacement (1); (d) Combination (1); (e) Displacement (1).	1+1+1+1+1
	OR: Litmus: HCl turns blue to red (1); NaOH and Na_2CO_3 turn red litmus blue (1); Add acid: Na_2CO_3 effervesces, NaOH does not (1); NaCl has no effect on litmus (1); Lime water confirms CO_2 (1).	1+1+1+1+1
27	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Metal oxides are basic (1); Hydrogen is displaced by reactive metals (1).	1+1+1+1+1

	OR: Tetravalency: four bonds (1); Catenation: long chains, branches, rings (1); Multiple bonds possible (1); Isomerism (1); Strong C-C bonds (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 10	3 x 3 + 1 = 10.
2	**(a)** C	Carbon takes oxygen from ZnO.
3	**(b)** thermal decomposition	Heat decomposes baking soda.
4	**(b)** +2	It loses two electrons.
5	**(c)** CH_3COOH	Oxidation.
6	**(a)** water contains 2 hydrogen atoms for each oxygen atom	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
7	**(a)** hydrogen	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$.
8	**(d)** is more reactive than iron	Zinc corrodes first.
9	**(d)** high energy is needed to remove 4 electrons	Too much energy.
10	**(c)** 3	$2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$.
11	**(a)** HCl and NaOH	Strong acid + strong base gives neutral salt.
12	**(a)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$	Only a more reactive metal displaces a less reactive.
13	**(b)** 6	4 C-H pairs plus 2 pairs in C=C.
14	**(d)** thermal decomposition	Heat causes the break-down.
15	**(a)** disappears	Soluble calcium hydrogencarbonate forms.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Adds quick lime or slaked lime (1); Basic substances neutralise the acid (1).	1+1
17	Ions are free in solution (1); Fixed in the solid lattice (1).	1+1
18	Add bromine water (1); Unsaturated decolourises it (1).	1+1
19	They decompose in the presence of light (1); $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (1).	1+1
20	Baking soda and tartaric acid release CO_2 (1); CO_2 bubbles make the cake rise (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Soap: fatty acid salts; detergent: sulphonate salts (1); Soap forms scum in hard water (1); Detergents don't (1).	1+1+1
22	Cathode: hydrogen; anode: oxygen (1); Volume ratio 2 : 1 (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1).	1+1+1
23	Acetic acid reacts with carbonate (1); $CaCO_3 + 2\,CH_3COOH \longrightarrow (CH_3COO)_2Ca + H_2O + CO_2$ (1); Bubbles of CO_2 (1).	1+1+1
24	Zinc layer isolates iron from air (1); Zinc is more reactive and gets oxidised first (1); So iron is protected even if scratched (1).	1+1+1
	OR: Sharing of electron pairs (1); Cl_2: single bond (1); O_2: double bond (1).	1+1+1
25	Balance N and H (1); $4\,NH_3 + 5\,O_2 \longrightarrow 4\,NO + 6\,H_2O$ (1); Oxidation (redox) of ammonia / combustion (1).	1+1+1
	OR: Mix equal amounts of the same concentration (1); H^+ and OH^- react to form water (1); Neutralisation gives pH 7 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Thermal: $CaCO_3 \longrightarrow CaO + CO_2$ (1); Electrolytic: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Photolytic: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (1); Photolytic used in photography (1); All need energy, so they are endothermic (1).	1+1+1+1+1

	OR: $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O$ (1); Heated at 373 K (1); $CaSO_4 \cdot \frac{1}{2}\,H_2O + 1.5\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$ (1); Uses: plaster, toys, decoration (1); Stored in moisture-proof container (1).	1+1+1+1+1
27	Bridges: painting, galvanisation (1); Aluminium: anodising (1); Ships: sacrificial zinc/magnesium plates (1); Zn, Mg are more reactive than iron (1); They corrode instead of iron (1).	1+1+1+1+1
	OR: $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (sunlight) (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (Ni) (1); $CH_2{=}CH_2 + Br_2 \longrightarrow CH_2Br{-}CH_2Br$ (1); Bromine water decolourised (1); Addition for unsaturated (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** X^-	It gains one electron.
2	**(d)** C_2H_4	Unsaturated compounds.
3	**(d)** prevent oxidation of the oil in the chips	Nitrogen is inert.
4	**(b)** NaOH	NaOH ionises completely in water.
5	**(b)** anodising	Anodising.
6	**(a)** 6 protons cannot hold 10 electrons easily	Nucleus can't hold extra electrons.
7	**(d)** 9	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$: 4+3+2=9.
8	**(c)** basic	Strong base and weak acid give basic salt.
9	**(d)** Mg > Zn > Fe > Cu	Standard reactivity series.
10	**(a)** 5	2 C-H plus 3 in triple bond.
11	**(b)** $CaO + H_2O \longrightarrow Ca(OH)_2$	Quick lime slaking releases heat.
12	**(d)** formation of gypsum	Water is taken up.
13	**(b)** $MgCl_2$	Mg^{2+} and Cl^-.
14	**(c)** addition reaction	Double bond opens.
15	**(d)** is more reactive than iron and forms a protective layer	Zinc protects the iron.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Water is polar and separates ions (1); Kerosene is non-polar (1).	1+1
17	$HCOOH + NaOH \longrightarrow HCOONa + H_2O$ (2).	2
18	It releases heat energy (1); $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O + \text{heat}$ (1).	1+1
19	pH 2 solution has more H^+ (1); 100 times as much (1).	1+1
20	Copper is below hydrogen in the series (1); Zinc is above hydrogen (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	A > C > B (1); A displaces hydrogen from acid and reacts with cold water (1); B does not react with acids, so it is least reactive (1).	1+1+1
22	Diamond has 3D arrangement (1); Graphite has layers (1); Free electrons in graphite make it conductor (1).	1+1+1
23	Blue colour fades; reddish-brown copper deposits (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); Zinc is more reactive than copper (1).	1+1+1
24	It forms gypsum (1); Gypsum is a hard interlocking solid (1); Setting takes place in a short time (1).	1+1+1
	OR: Metals from reduction contain impurities (1); Electrolytic refining: impure Cu anode and pure Cu cathode (1); Pure copper deposits on the cathode (1).	1+1+1
25	Hydrophobic tail avoids water (1); Heads face water (1); Cluster forms a sphere (micelle) (1).	1+1+1
	OR: Crackling sound (1); Brown fumes of NO_2 (1); Yellow residue of PbO (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Oxidised: H_2; reduced: CuO (1); (i) Oxidising agent CuO; reducing agent H_2 (1); (ii) Oxidised: CO (1); (ii) Reduced: Fe_2O_3 (1); (ii) Oxidising agent Fe_2O_3; reducing agent CO (1).	1+1+1+1+1

	OR: $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $NaOH + HCl \longrightarrow NaCl + H_2O$ (1); Bases with Zn: $2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$ (1); Acids and bases both conduct in solution (1).	1+1+1+1+1
27	Hg: heating cinnabar, as unreactive (1); $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$ (1); Zn: calcination and reduction with carbon (1); $ZnO + C \longrightarrow Zn + CO$ (1); Na: electrolysis of molten NaCl (1).	1+1+1+1+1
	OR: C_2H_6, C_2H_4, C_2H_2 (1); Single, double, triple bonds (1); Ethane clean flame (1); Ethene and ethyne sooty (1); Ethene and ethyne show addition (1).	1+1+1+1+1

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